SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA1021	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 1 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:	RASHMITHA SENTHIL KUMAR	**Reg. No.:**	192525034
Date:	31/10/2026	**Duration:**	60 minutes

Code of Conduct

*I **RASHMITHA SENTHIL KUMAR (192525034)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **RASHMITHA. S**

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario** ○ Carefully understand the given problem statement.
    - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements** ○ List all functional requirements required by the proposed system.
    - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases** ○ Identify all primary and secondary actors interacting with the system. ○ List the major use cases associated with each actor.
    - Draw a Use Case Diagram representing the interactions between actors and the system.

- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential

(15%)	sections.			components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

REAL-TIME SMART MUNICIPAL AQUEDUCT LEAK DETECTION & AUTOMATED VALVE ISOLATION FABRIC

1. Industry Context and Problem Interpretation:

Municipal water distribution networks are geographically dispersed cyber-physical systems connecting treatment plants, reservoirs, pumps, pressure zones, pipelines and consumer connections. Hidden underground leaks waste treated water and pumping energy, while major pipe bursts can erode road foundations, flood properties and interrupt service. Conventional monitoring through periodic inspection and customer complaints can leave a significant delay between fault onset and intervention.

The engineering problem is to monitor thousands of utility nodes concurrently, detect pressure and flow anomalies, localize probable leaks from acoustic travel-time differences, and coordinate safe valve isolation. The system must also adapt pressure setpoints to changing demand and retain trustworthy utilization and command records for utility oversight. A pressure drop alone does not prove leakage: firefighting demand, pump starts, scheduled switching, pipe transients and sensor defects can generate similar signals.

Core engineering constraints:

The design must account for limited sensor power, intermittent field connectivity, inconsistent timestamps, variable acoustic propagation speed, uncertain pipe topology, valve actuator limits and the public-safety consequences of an incorrect closure. The phrase “zero data loss” is treated as a target requiring explicit assumptions about local storage capacity and failure domains, not as an unconditional physical guarantee. Likewise, “near-zero flooding risk” cannot be guaranteed solely through automation.

2. Objectives, Scope and Success Criteria:

The four scenario objectives are: durable telemetry ingestion; acoustic leak localization; verified automated isolation; and dynamic pressure balancing. A fifth cross-cutting requirement is audit-ready, tamper-evident utilization records. The proposed prototype should use simulated sensors and simulated valve actuators; production integration requires additional hydraulic, control-system and cybersecurity certification appropriate to the municipality.

2.1 Functional Requirements:

Functional requirements describe the operations and services that the proposed Municipal Aqueduct Leak Detection System must perform.

- FR-01 – Real-Time Sensor Data Acquisition: The system shall continuously collect pressure, flow, acoustic, soil moisture and valve-status measurements from distributed municipal water network sensors.

- FR-02 – Sensor Data Validation: The system shall validate incoming telemetry using sensor identifiers, timestamps, sequence numbers, calibration metadata and signal-quality indicators.

- FR-03 – Automated Leak Detection: The system shall analyse pressure changes, flow imbalances and acoustic measurements to identify suspected leaks and major pipeline bursts.

- FR-04 – Acoustic Leak Localization: The system shall estimate the location of underground leakage using the time difference between acoustic signals detected by sensors installed along a pipeline.

- FR-05 – Automated Valve Isolation: The system shall generate isolation commands for verified critical leaks only after authorization and safety-interlock requirements are satisfied.

- FR-06 – Dynamic Pressure Management: The system shall monitor water pressure across distribution zones and recommend or execute validated pressure adjustments based on changing demand.

- FR-07 – Real-Time Monitoring Dashboard: The system shall display sensor measurements, pressure trends, detected anomalies, estimated leak locations and valve-operation status.

- FR-08 – Alert and Notification Management: The system shall notify authorized utility operators when critical leakage, sensor malfunction, communication failure or unsuccessful valve isolation occurs.

- FR-09 – Fault Detection and Recovery: The system shall identify unavailable or unreliable sensors and support recovery of buffered telemetry following temporary communication interruptions.

- FR-10 – Audit Logging: The system shall maintain tamper-evident records of important monitoring events, leak-detection decisions, operator actions and simulated valve operations.

2.2 Non-Functional Requirements:

Non-functional requirements define the quality attributes and operating constraints of the proposed system.

NFR-01 – Performance: The system shall support continuous telemetry processing and provide timely detection and alert generation. Numerical latency limits shall be established during requirements validation.

NFR-02 – Scalability: The architecture shall support thousands of distributed sensing nodes through scalable communication and processing components.

NFR-03 – Reliability: The system shall detect communication failures and support durable buffering and replay within defined storage and failure limits.

NFR-04 – Security: The system shall enforce authenticated access, role-based authorization, encrypted communications and protected control interfaces.

NFR-05 – Availability: Monitoring services shall support fault recovery and redundancy according to availability targets agreed with the municipal utility.

NFR-06 – Usability: The monitoring dashboard shall provide understandable sensor readings, incident classifications, warning messages and valve-status information.

NFR-07 – Maintainability: The software shall use modular components, documented interfaces and testable processing functions to simplify updates and fault diagnosis.

NFR-08 – Safety: The system shall prevent unverified leak conditions from automatically triggering unsafe valve operations.

NFR-09 – Data Integrity: The system shall identify duplicate events, preserve required telemetry records and detect unauthorized modification of protected audit records.

NFR-10 – Interoperability: The architecture shall support integration with suitable IoT communication interfaces, databases and industrial supervisory control systems.

3. Proposed Distributed System Architecture:

A layered architecture separates low-power field devices from cloud or datacenter analytics and isolates safety-critical actuation from general-purpose services. Pressure transducers, flow meters, acoustic pickups, soil-moisture probes and valve-position feedback devices feed local gateways. Gateways timestamp, validate, filter and persist messages. A durable message broker fans telemetry out to detection, topology, storage and dashboard services.

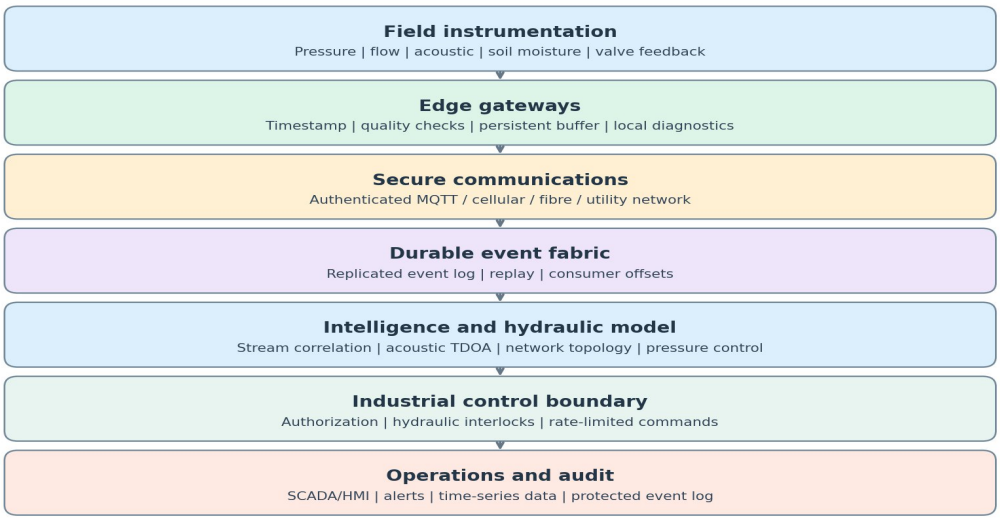


Figure 1. Conceptual end-to-end aqueduct monitoring architecture.

Data-plane and control-plane separation:

The data plane carries measurements and diagnostic events; the control plane handles policy, topology, operator roles, setpoint changes and valve commands. Control commands must never be accepted directly from untrusted telemetry topics. A command should include a unique request identifier, target valve, requested operation, preconditions, validity interval, authorizing identity and expected feedback. Duplicate commands should be rejected or treated idempotently.

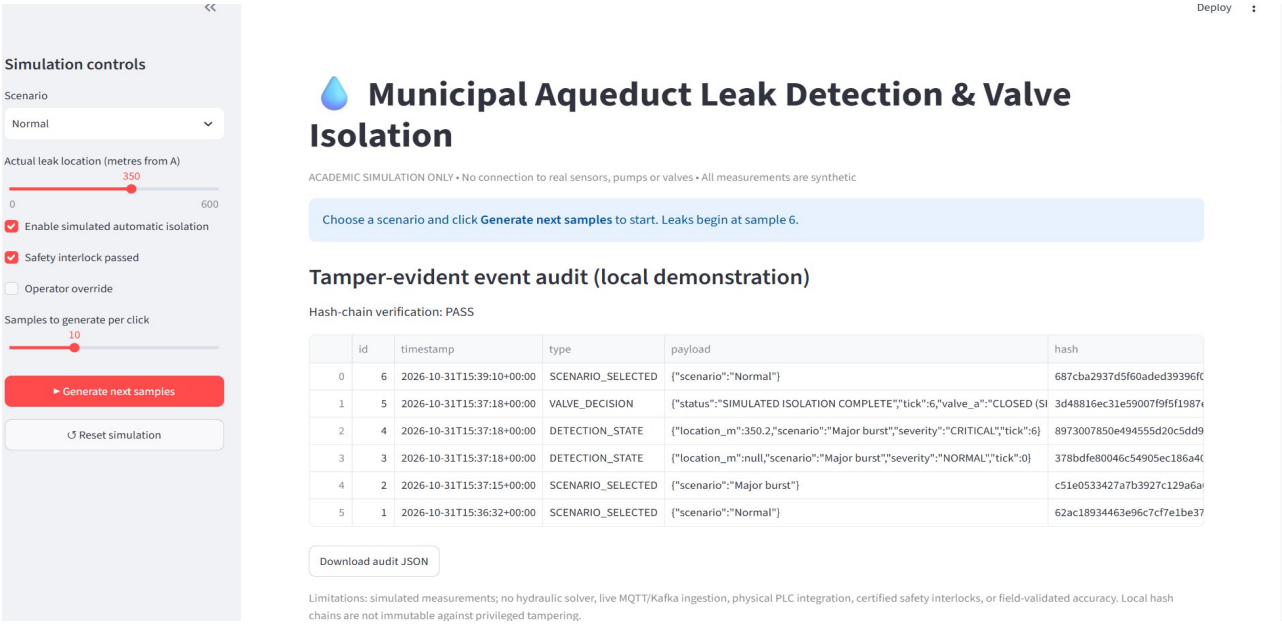


Figure 2: Slow Leak Detection and Pressure Monitoring

3.1 Actor Identification and Use Case Modelling:

The proposed system interacts with human operators, maintenance personnel, administrators and external field devices. Identifying these actors helps define system boundaries, responsibilities and functional interactions.

Primary Actors:

- Municipal Utility Operator: Monitors water distribution conditions, reviews alerts and authorizes operational actions.
- Maintenance Engineer: Examines suspected leak locations, investigates pipeline failures and updates maintenance status.
- System Administrator: Manages user access, device configurations, monitoring policies and system maintenance.

Secondary Actors:

- IoT Sensor Gateway: Supplies telemetry measurements collected from distributed water sensors.
- Valve Control System: Receives authorized isolation commands and returns valve-status feedback.

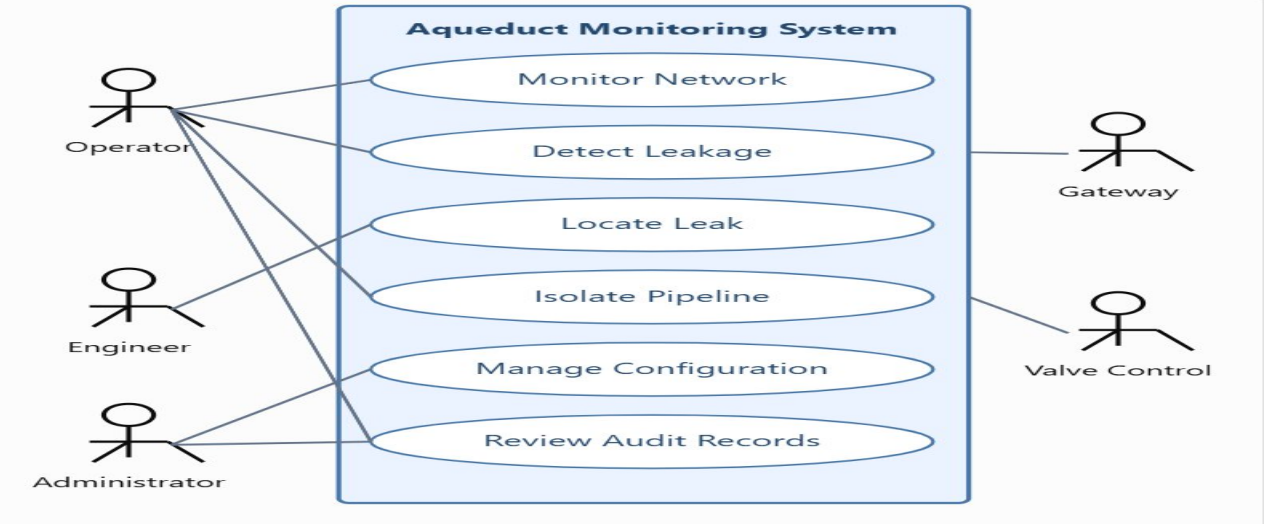


Figure 3: UML Use Case Diagram for the Municipal Aqueduct Leak Detection System. Conceptual diagram showing major actor interactions.

3.2 Major Use Cases:

UC-01 – Monitor Water Network: The utility operator views current pressure, flow and sensor-health information through the dashboard.

UC-02 – Detect Pipeline Leakage: The system analyzes sensor measurements to identify suspected or verified pipeline leakage.

UC-03 – Locate Underground Leak: The system estimates the leak position using acoustic time-difference analysis.

UC-04 – Manage Leak Alerts: The utility operator reviews detected anomalies and their associated evidence.

UC-05 – Isolate Damaged Pipeline: The system requests authorized valve isolation after verifying leak severity and safety conditions.

UC-06 – Inspect and Repair Pipeline: The maintenance engineer uses the estimated location and incident details to investigate the damaged section.

UC-07 – Manage System Configuration: The administrator manages device registration, access permissions and system configuration.

UC-08 – Review Audit Records: Authorized users inspect historical monitoring events, valve decisions and integrity-verification results.

3.3 Use Case Description – Automated Valve Isolation:

Use Case ID: UC-05

Use Case Name: Automated Valve Isolation

Primary Actor: Municipal Utility Operator

Supporting Actor: Valve Control System

Preconditions: A critical leak has been verified, the affected segment has been identified, and the required authorization and safety conditions are satisfied.

Main Flow:

1. The system identifies a verified critical pipeline leak.
2. The isolation planner identifies candidate valves.

3. The system evaluates operating constraints and safety interlocks.
4. The authorized control interface issues a simulated valve command.
5. The valve simulator updates the valve state.
6. The system records the command, decision and resulting status.
7. The dashboard displays the isolation outcome.

Alternative Flow: If the leak remains unverified or a safety condition fails, automatic isolation is blocked and an alert is generated.

Postcondition: The simulated valve state and isolation decision are recorded for monitoring and audit purposes.

4. Sensor Placement and Data Acquisition:

Pressure sensors are most useful near pressure-zone boundaries, pumps, critical junctions and historically fragile segments. Flow meters at district-metered area inlets and outlets help estimate water balance.

Acoustic sensors should bracket candidate pipe runs where geometry and attenuation permit reliable correlation. Soil moisture can support diagnosis but may also respond to rainfall or irrigation. Valve limit switches and actuator telemetry are required to confirm physical movement rather than infer it from command acknowledgement.

Each measurement should include sensor_id, event_id, UTC timestamp, sequence_number, pipe_segment_id, value, unit, calibration_version, signal_quality and gateway_id.

The system must maintain metadata for pipe material, diameter, wall thickness, sensor spacing and calibration history.

Time synchronization quality is especially important for acoustic time-difference methods; where sampling rates are high, correlated waveform acquisition may require synchronized local recorders rather than independent low-rate IoT messages.

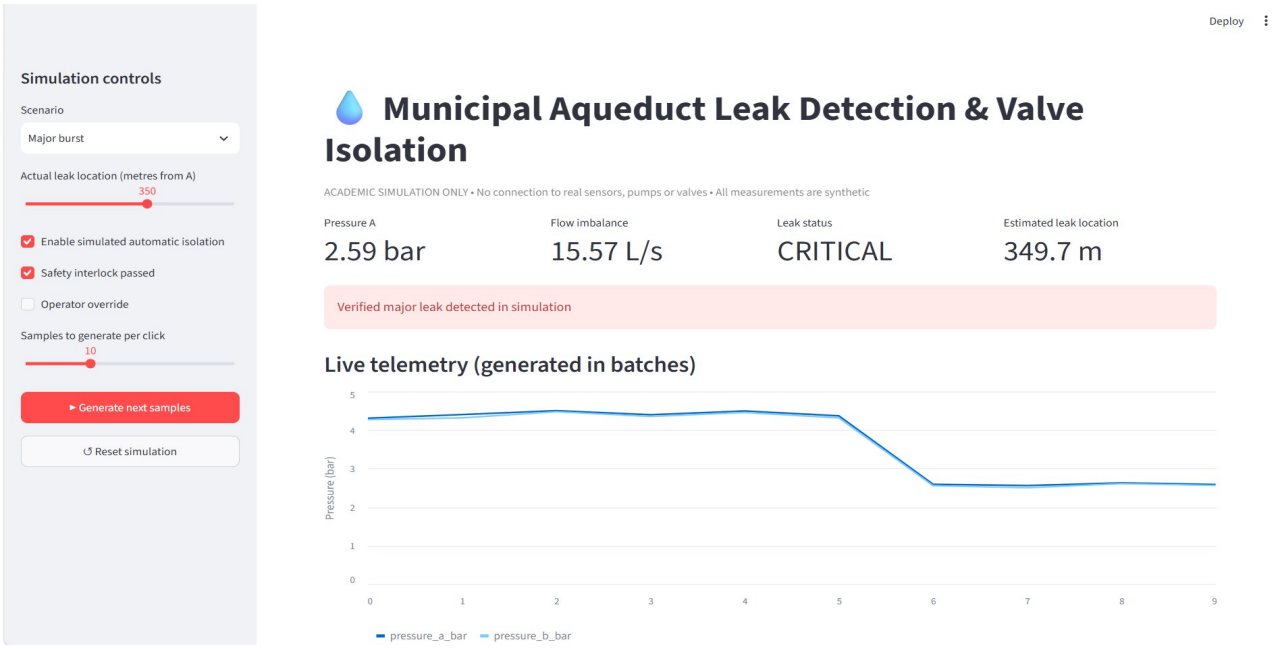


Figure 4: Major Pipeline Burst Detection and Estimated Leak Location

Sampling and energy trade-offs:

Routine pressure and flow monitoring can use periodic samples, with faster event-triggered acquisition during anomalies. Acoustic waveform capture can be processed locally into correlation windows or features to reduce uplink bandwidth. Exact sampling rates are not specified by the scenario and must be selected from leak-size sensitivity, pipe characteristics, device power budget and communication constraints.

4.1 Software Requirement Specification:

The Software Requirement Specification (SRS) defines the scope, data requirements, external interfaces and major features of the proposed Municipal Aqueduct Leak Detection System. It establishes a structured basis for software design, implementation and validation.

4.2 Scope of the System:

The proposed system is designed to monitor municipal water distribution networks, detect pipeline anomalies, estimate leak locations and coordinate safety-validated valve isolation. The system also supports pressure-management decisions, incident monitoring and audit-record generation.

The academic prototype is limited to simulated sensor readings, software-based leak detection, acoustic localization calculations, simulated valve decisions, dashboard visualization and local

audit logging. Physical water sensors, industrial valve actuators and production SCADA integration are outside the implemented prototype scope.

4.3 Data Requirements:

The system requires pressure readings, water flow rates, acoustic time-difference measurements, soil moisture indicators, sensor identifiers, timestamps, pipeline segment details and valve states.

Each telemetry event should include an event identifier and quality information to support validation and traceability. Historical records should preserve detected anomalies, estimated leak positions, command decisions and relevant operating conditions.

4.4 External Interface Requirements:

User Interface: A dashboard shall display network monitoring information, sensor measurements, leak classifications, location estimates and valve-status information.

Sensor Interface: The proposed architecture shall accept measurements from distributed field devices through authenticated gateways and suitable communication protocols.

Control Interface: The proposed control layer shall communicate authorized valve commands through validated control interfaces and receive execution feedback.

Database Interface: The system shall support storage and retrieval of telemetry, incident history and audit records.

4.5 System Features:

The principal system features include real-time telemetry monitoring, multi-sensor leak detection, acoustic leak localization, simulated valve isolation, safety-interlock verification, pressure monitoring, sensor-fault identification and tamper-evident event logging.

These features are intended to improve incident visibility, support faster maintenance decisions and provide a foundation for future integration with municipal water infrastructure.

5. Durable High-Velocity Streaming and Recovery:

A lightweight field protocol such as MQTT can transport gateway data to authenticated ingress services. MQTT delivery quality-of-service settings improve transport behavior but do not alone

guarantee end-to-end persistence. The ingress service should validate schemas, persist events to a replicated log and acknowledge only after the required durability condition is satisfied. Partitioning by district or pipe segment preserves useful local ordering while distributing workload across processors.

Gateways should use bounded persistent queues with disk-health checks, retry policies and backlog alarms. Consumers must tolerate late and duplicate events using stable event identifiers, event-time windows, checkpoints and idempotent writes. If local buffer capacity is exhausted or multiple storage replicas fail, data loss remains possible; these limits must be explicitly monitored and reported.

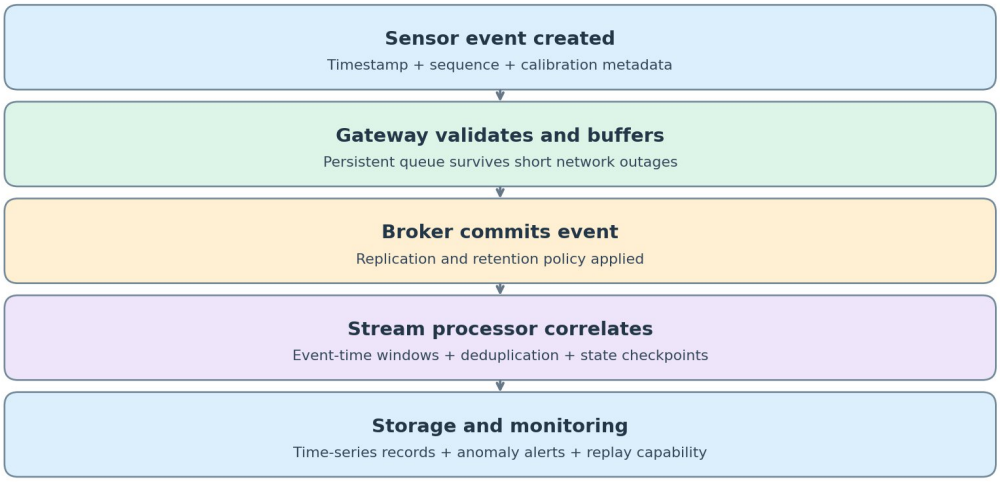


Figure 5. Durable ingestion, processing and replay workflow.

Illustrative load estimation:

If 5,000 devices each send one 200-byte message per second, the raw payload rate is approximately 1 MB/s, or 86.4 GB/day before transport overhead, replication, indexes and retention. This is a sizing illustration, not a measured deployment workload. Bursty acoustic data may dominate bandwidth and requires separate capacity planning.

6. Multi-Sensor Leak Detection and False-Alarm Reduction:

The detection service should first establish normal behavior using time-of-day demand, pump schedules, zone boundaries and weather or operational context. Candidate anomalies include rapid negative pressure gradients, unexplained increases in district flow, persistent minimum-night-flow

elevation, correlated acoustic energy and moisture anomalies. A verified leak should require a documented combination of evidence, confidence thresholds and persistence rules rather than a single threshold crossing.

A staged decision process can distinguish suspected leak, corroborated leak, confirmed burst, isolation authorization and recovery. The platform should store the evidence that caused each state transition. False-positive analysis must include demand spikes, pump trips, hydrant use, planned valve operations, sensor drift and communication timestamp errors.

7. Acoustic Time-Difference Leak Localization:

A leak produces turbulence and pipe-wall vibration that can propagate in both directions along a pipe. For two sensors A and B separated by distance L, let x be the leak distance from A and c the calibrated propagation speed. If $t_A=x/c$ and $t_B=(L-x)/c$, the arrival-time difference $\Delta t=t_B-t_A=(L-2x)/c$. Rearrangement gives $x=(L-c\Delta t)/2$. This sign convention is essential: swapping the order of the time difference reverses the sign. Correlation lag must be defined consistently in software.

Illustrative example: L=600 m, c=1,000 m/s and $\Delta t=-0.10$ s, giving $x=(600-(1{,}000\times-0.10))/2=350$ m from A and 250 m from B. These values are hypothetical. Real measurements require propagation-speed calibration, sufficient waveform similarity, sensor synchronization and treatment of reflections, branches and attenuation.

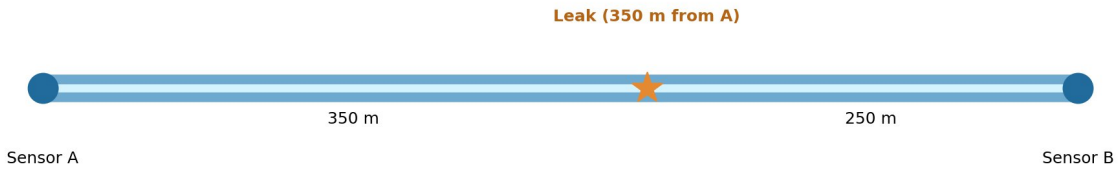


Figure 6. Illustrative acoustic sensor geometry and leak position.

Localization uncertainty and validation:

Cross-correlation estimates the lag between synchronized acoustic records. The estimated location should include a confidence interval reflecting timing uncertainty, propagation-speed uncertainty and pipe geometry. Test with multiple leak positions, pipe materials, signal-to-noise ratios and

background vibration levels. Report mean absolute error, worst-case error and percentage of events for which localization is unavailable rather than only the best example.

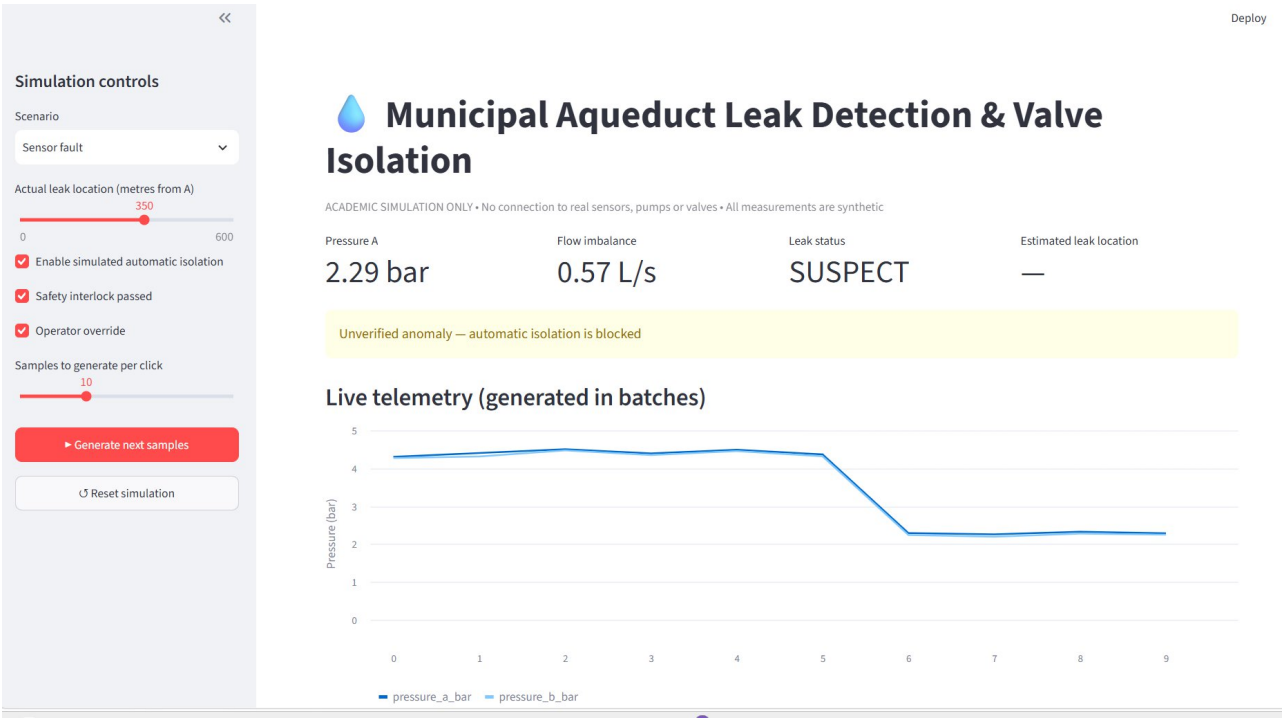


Figure 7: Sensor Fault Detection and Unverified Anomaly Handling

8. Hydraulic Network Model and Safe Valve Isolation:

The hydraulic topology service should represent junctions, pipes, reservoirs, pumps, pressure zones, valves and critical service points as a connected graph. A candidate leak is mapped to one or more pipe segments. The isolation planner identifies a feasible valve cut-set that disconnects the damaged segment while minimizing affected customers, but topology alone is insufficient: each candidate action must also be checked with hydraulic simulation or validated operating rules.

Before actuation, the safety boundary checks leak evidence, valve accessibility, current state, downstream supply, minimum service pressure, allowable transient rate, command age, operator policy and emergency overrides. Commands should be sequenced to avoid abrupt closure and water hammer. Actual closure speed and allowable pressure transients must be determined by qualified hydraulic engineers. The platform should require positive position feedback and measured post-operation pressures, and escalate if a valve sticks or feedback is inconsistent.

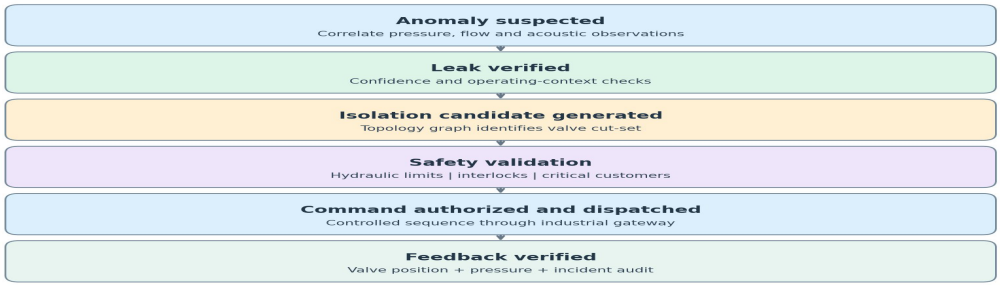


Figure 8: Safety-gated automated isolation sequence.

Failure and fallback states:

If confidence is inadequate, the system should alarm and request investigation rather than close valves. If a candidate isolation would violate service constraints, it must be rejected or escalated. If communication is lost during execution, local industrial controllers must follow an independently validated safe-state policy. The supervisory system should reconcile actual valve state after reconnection; a network timeout must never be treated as proof that a valve closed.



Figure 9: Tamper-Evident Event Audit and Hash-Chain Verification

9. Dynamic Pressure Balancing and Optimization:

Demand falls and rises by hour, zone and season. Pressure management should reduce excessive pressure without violating minimum consumer supply requirements. A supervisory optimizer can

use demand forecasts, pressure observations, pump curves, tank levels and a calibrated hydraulic model to propose setpoints. The objective may combine pressure deviation, pumping energy, leakage-related pressure exposure and actuator movement penalties, subject to constraints on minimum service pressure, maximum pressure, pump operation and valve movement rate.

A model-predictive approach can compute a short-horizon schedule and apply only the first validated action before re-evaluating measurements. A simpler rule-based controller may be more appropriate for an initial prototype. All automated setpoint changes should be bounded, reversible where safe, logged and monitored for oscillation. The assignment provides no numerical pressure targets, so values must be defined by the utility rather than invented.

10. Fault Tolerance, Availability and Disaster Recovery:

Fault tolerance spans field sensors, gateways, communications, streaming services, databases and actuators. Replicated message storage and stream checkpoints allow recovery after process crashes. Local persistent queues protect against temporary network outages. Redundant sensors improve observability where critical, but must not be assumed to provide independent evidence if they share a common power source or communication path.

Table 1. Selected fault scenarios and proposed responses

Failure	Safe system response
Gateway disconnect	Buffer within capacity; report stale telemetry
Sensor disagreement	Lower confidence; flag calibration or health issue
Broker node crash	Fail over and replay committed messages
Actuator stuck	Stop unsafe sequence; raise high-priority alarm
Central analytics outage	Retain independent local controller safety rules

11. Security, Privacy and Audit-Ready Utilization Records:

Threats include spoofed sensor data, compromised gateway firmware, replayed messages, unauthorized setpoint changes, stolen operator credentials, malicious valve commands and audit-log alteration. The design should apply mutual authentication where feasible, encryption in transit,

secure firmware updates, network segmentation, least-privilege role-based access, multi-factor authentication for operators and separated management and operational networks.

Audit events should include an event identifier, trusted timestamp, actor identity, asset identifier, prior state, requested action, safety-check result, final state and linked evidence. Append-only storage and cryptographic hash chaining can make unauthorized changes detectable; protected signing keys, independent replicas and retention governance are required for stronger integrity. “Unalterable” should be interpreted as a requirement for defensible tamper resistance and evidence, not a guarantee that no administrator or physical attacker could ever destroy records.

12. Proposed Software Stack and Prototype Modules:

A feasible academic prototype can use Python for telemetry and acoustic simulation, MQTT for gateway messaging, Kafka for event retention, a Python processing service for correlation, NetworkX for graph-based pipe topology, EPANET for hydraulic scenarios, PostgreSQL/TimescaleDB for records, and a web dashboard for incident visualization. These are proposed choices, not dependencies mandated by the assignment. A smaller initial implementation can use Python, MQTT, SQLite/PostgreSQL and a simple dashboard before adding distributed streaming infrastructure.

Module breakdown:

- Module A — Synthetic telemetry generator: produce normal demand, slow-leak, burst, sensor-failure and packet-delay scenarios.
- Module B — Event ingestion and recovery: validate messages, persist event identifiers, deduplicate and replay buffered data.
- Module C — Detection and localization: correlate sensors, classify incidents and estimate acoustic time delay.
- Module D — Hydraulic safety and valve simulator: compute candidate isolation plans and reject unsafe operations.
- Module E — Monitoring and audit: display event history, valve states, zone pressures and integrity checks.

13. Algorithm and Processing Logic:

Algorithm 1 — Verified leak detection:

Input: pressure P(t), flow Q(t), acoustic features A(t), valve and pump states, topology and sensor quality. Step 1: reject or flag stale and low-quality readings. Step 2: compute pressure gradients and flow residuals against an operating baseline. Step 3: identify candidate segments using correlated sensor anomalies. Step 4: estimate acoustic time lag and location when valid paired waveforms exist. Step 5: score evidence and rule out known operational transitions. Step 6: classify the incident and store the evidence. Step 7: request a safety-checked isolation plan only when the approved confidence and severity criteria are met.

Algorithm 2 — Isolation decision:

Input: confirmed incident, network graph, valve states, critical consumer constraints and hydraulic model. Step 1: enumerate candidate valves around the affected segment. Step 2: construct candidate isolation sets. Step 3: simulate supply continuity and pressure consequences. Step 4: discard candidates that violate safety constraints. Step 5: select the permitted option with the least operational disruption. Step 6: require applicable authorization and issue a rate-limited command sequence. Step 7: verify physical valve feedback and pressure response. Step 8: record Pass, Fail or Warning with evidence and operator notification.

14. Quantitative Evaluation and Test Plan:

Evaluation must compare proposed behavior with a defined baseline using repeatable synthetic datasets. Detection precision is TP/(TP+FP), recall is TP/(TP+FN), and false-alarm rate should be measured over a stated operating duration. Leak-location error is |estimated distance−actual distance|. Ingestion durability is assessed by reconciling generated event IDs with persisted IDs after fault injection.

End-to-end response latency should be broken into detection, decision, authorization, dispatch and physical actuator phases.

Table 2. Module-level test and validation matrix

Test	Observed evidence	Status
Major-burst detection	CRITICAL status and pressure drop displayed	Pass
Acoustic location calculation	349.7 m estimated versus 350 m configured	Pass
Slow-leak handling	SUSPECT status; isolation blocked	Pass
Sensor-fault handling	SUSPECT status without a location estimate	Pass
Simulated valve isolation	Audit record reports isolation completed	Pass*
Audit-chain verification	Dashboard displays hash-chain PASS	Pass*
Safety-interlock rejection	Separate failed-interlock test not shown	Not tested
Gateway outage and replay	No execution evidence provided	Not tested
Dynamic pressure control	No controller validation provided	Not tested

Pass / Fail / Warning criteria:

Pass means all agreed acceptance criteria are met and supported by recorded measurements. Fail means a required threshold or safety constraint is breached. Warning means the result is inconclusive because evidence, calibration or repeatability is insufficient. No test in this document is marked Pass because the prototype has not been executed.

14.1 Assumptions:

The proposed system is designed under the assumption that municipal pipelines can be represented using an identifiable network topology and that sensing devices can be associated with known pipeline segments.

It is assumed that pressure, flow and acoustic measurements are available with sufficient quality for anomaly detection. Acoustic propagation speed is assumed to be calibrated or estimated for the monitored pipe segment.

The academic prototype assumes synthetic sensor data and software-simulated valve operations. It does not assume that simulated results establish performance under real-world municipal operating conditions.

14.2 Constraints:

Technical Constraints: Sensor accuracy, limited battery capacity, communication delays, acoustic interference, uncertain pipe characteristics and processing capacity may affect system performance.

Operational Constraints: Valve isolation must respect minimum service pressure, critical water-supply requirements, authorized operating procedures and safety interlocks.

Economic Constraints: Deployment cost depends on sensor density, communication infrastructure, gateway equipment, storage requirements and maintenance resources.

Legal and Safety Constraints: Real municipal deployment must comply with applicable water utility regulations, cybersecurity requirements, operational safety policies and relevant infrastructure standards.

Time Constraints: Development and testing activities must be completed within the academic project schedule. Production deployment would require additional engineering validation and approval.

14.3 Requirement Completeness and Consistency Validation:

Requirement validation ensures that the proposed system specification addresses the scenario objectives without ambiguity, unnecessary duplication or contradictory behaviour.

Completeness: The requirements cover telemetry acquisition, leak detection, localization, valve isolation, pressure management, fault handling, dashboard monitoring and audit logging.

Consistency: Automated valve isolation is permitted only when leak-verification and safety conditions are satisfied. This prevents conflict between rapid response and operational safety.

Unambiguity: Terms such as real-time, high availability, zero data loss and safe pressure must be associated with measurable acceptance criteria before production deployment.

Traceability: Each functional requirement should be linked to a corresponding use case, system component and verification activity.

Feasibility: The academic prototype demonstrates selected functions through software simulation, while full distributed ingestion, physical valve control and hydraulic optimization remain proposed capabilities.

14.4 Suggested Improvements:

The requirement specification can be strengthened by defining measurable performance thresholds, documenting stakeholder approval, introducing a formal requirements traceability matrix and validating the architecture against representative hydraulic network conditions.

Future development should also include communication failure testing, physical actuator feedback, independent safety assessment and field calibration.

15. Implementation Requirements and Screenshot Evidence:

The scenario explicitly requests design and development, so an executable prototype is strongly advisable for demonstrating the engineering claims. The reference document provides an analytical writing pattern but does not specify a compulsory implementation tool, a marking rubric, or a mandatory screenshot count for this scenario. Therefore, implementation is recommended rather than represented as an externally confirmed submission requirement.

For a credible submission, capture real screenshots after executing the prototype: (1) sensor data generator and sample event stream; (2) dashboard with normal pressure trends; (3) simulated burst and correlated leak alert; (4) acoustic localization result with known ground truth; (5) safe valve isolation and position feedback; (6) rejected unsafe command; (7) simulated communication outage and replay; and (8) tamper-evident audit verification. Never submit invented screenshots as experimental evidence.

16. Expected Outcomes, Limitations and Engineering Trade-Offs:

Expected benefits include earlier discovery of underground failures, reduced duration of uncontrolled water discharge, more focused maintenance dispatch, better pressure stability and more traceable utility reporting. Actual water savings, flood-risk reduction and cost benefits require comparison with field baseline measurements. Detection may be unreliable on noisy, branched or

poorly instrumented pipes; acoustic propagation can vary significantly; a hydraulically correct valve plan may still be infeasible because of actuator or communication faults.

The design trades communication and storage costs against sensing fidelity, and automation speed against public-safety assurance. The strongest architecture uses local resilience, independent safety controls, multi-sensor verification and measurable auditability rather than assuming a central AI service can safely command all field devices without constraints.

17. Conclusion:

The proposed Real-Time Smart Municipal Aqueduct Leak Detection & Automated Valve Isolation Fabric integrates distributed telemetry ingestion, acoustic time-difference localization, hydraulic network modeling, safety-gated valve coordination, dynamic pressure management and tamper-evident records. Its main engineering contribution is the controlled transition from sensor evidence to explainable, verified operational action across a geographically distributed water network. A simulation-first prototype can evaluate detection quality, localization accuracy, event durability and interlock behavior before any real-world deployment. Claims of zero data loss, near-zero flooding or automatic safety cannot be validated without explicit failure assumptions and field evidence.